# Nermine Moumen

**Final-year Engineering Student in Industrial Chemistry**

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## EDUCATION

**National Institute of Applied Sciences and Technology (INSAT)** *2022–Present*

Industrial Chemical Engineering student, specialized in Process and Environment, with a strong foundation in electrochemical kinetics, electrochemical measurements and analytical chemistry. Interested in electrochemical methods applied to water quality and in the interpretation of electrochemical behavior. Seeking a 6-month internship starting February 2027.

## PROFESSIONAL EXPERIENCE

**SEGULA Technologies – Electrochemical Battery Modeling Internship** *06/2026–08/2026*

- Modeled a **Li-ion cell** (NMC811/Graphite-SiOx) using the Single Particle Model (SPM) implemented in MATLAB.
- Implemented the electrodes' **open-circuit potential (OCP) curves** and solved the PDEs with **FVM**.
- Optimized the mesh and diffusion coefficient values, reducing the RMSE error **from 202.13 mV to 40.5 mV**.
- Extracted and processed **experimental discharge data** from a database to validate MATLAB and COMSOL results.

**Control Union – Quality Control Internship** *08/2025–09/2025*

- Performed **laboratory analysis** of lubricants (viscosity, heavy metals by **ICP-OES**) following standardized procedures.
- Prepared and interpreted **analytical reports** to verify compliance with ASTM D5185 and **ISO 17025** standards.

**ETAP – Water Treatment and Petrochemistry Internship** *07/2024–08/2024*

- Performed physico-chemical and ionic analyses of **water samples** (pH, salinity, chlorides via **ion chromatography**).
- Studied water treatment methods **(coagulation-flocculation, ion exchange)** for contaminant and **corrosion** control.
- Performed organic geochemistry techniques **(GC, GC-MS, Rock-Eval pyrolysis)** for hydrocarbon **characterization**.

## ADDITIONAL EXPERIENCE

**Cascade – Online Chemistry Teacher (Part-time)** *2024–Present*

- Provide individualized online chemistry tutoring for French Terminale students, covering exam-level concepts.

## ACADEMIC PROJECTS

**Electrochemical Treatment of an Industrial Effluent by Electroflocculation**

- Carried out **electrochemical measurements** on aluminum **electrodes**: current-potential curves and **pulse polarography**.
- Calculated the **faradaic yield** with **Faraday's law**, linking electrode dissolution to the charge passed.
- Removal of $Cd^{2+}$ and $Zn^{2+}$ from a simulated effluent: **85% and 57%** in 20 min.

**Molecularly Imprinted Polymers (MIPs) for Pharmaceutical Applications**

- Synthesis and characterization of a sulpiride-selective MIP via radical polymerization, with **Soxhlet extraction**, **UV-Vis** spectrophotometry monitoring, and validation through controlled-release kinetics (240 min).
- Synthesis of a **core-shell MOF/polymer MIP** for targeted doxorubicin delivery, achieving 85% encapsulation efficiency.

**PEM Fuel Cell and Electrolyzer Modeling with COMSOL and ANSYS Fluent**

- Modeling of fuel cells (two-phase flow, **thermal effects**) at **imposed current** then at **imposed potential**.
- Modeling of a 3D PEM electrolyzer for **hydrogen production**: charge conservation, identification of the OER.

**Prediction of Liquid–Liquid Equilibria and Cross-flow and Countercurrent Extraction for Ternary Systems**

- Formulated the LLE thermodynamic system and solved it via Rachford-Rice and **arc-length continuation**.
- Reproduced 7 Mathematica notebooks proving equivalence of the **Hunter-Nash** and **McCabe-Thiele** methods.
- Reimplemented workflow in **Python**, MATLAB, and Mathematica (spline/fsolve, SciPy/mpltern, Interpolation/FindRoot).

## TECHNICAL SKILLS

- **Lab Techniques:** Chromatography (HPLC, GC, TLC), Spectrometry (MS, ICP-OES), Spectroscopy (NMR, FTIR, UV-Vis), Diffraction (XRD), Microscopy (SEM, TEM), Electrochemistry (Cyclic voltammetry, Polarography).
- **Simulation:** Aspen Plus, Aspen HYSYS, COMSOL Multiphysics • **Programming:** *MATLAB, Mathematica, Python*
- **Data & Office:** Microsoft Office Suite, OpenLCA • **Standards:** ISO 9001, ISO 14001, ISO 45001, ISO 17025, ISO 14040

## EXTRACURRICULAR ACTIVITIES

- **IEEE INSAT** – Chair of EMBS; Organizing Chair of TRSYP 3.0; Social Activities Chair of MENASYP.
- **THEATRO INSAT** – Member of the dance team.

## LANGUAGES AND CERTIFICATES

**Languages**

English (Fluent), French (Fluent), Arabic (Native).

**Certificates**

Aspen HYSYS – MATLAB